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Rössler Sections and Historical Cast

Return maps in a chaotic flow

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Math Sciences

MA4535 — Nonlinear Dynamics & Chaos | Lecture 8 | September 18, 2026

Rössler Flow

- For $\boldsymbol{y} = (y_1, y_2, y_3)$, use

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}) = \begin{pmatrix} -y_2 - y_3 \\ y_1 + a y_2 \\ b + y_3(y_1 - c) \end{pmatrix}, \qquad a = b = 0.2, \quad c = 5.7.$$

- The first two coordinates rotate. The third lifts and folds each trajectory.
- Typical attractor trajectories do not repeat exactly [13].

Rössler Flow in Words

- When $y_3 \approx 0$, the top two equations approximate an outward spiral (y_1, y_2).
- When $y_1 > c$ and $y_3 > 0$, y_3 grows. The $-y_3$ term then pushes y_1 below c.
- Rössler described the return map this way [13]

  "The flow in state space allows for a 'folded' Poincaré map (horseshoe map)."

- Trajectories stretch and fold as they go over the hump. Trajectories never cross.

Plane Section $y_1 = 0$

- Set $h(\boldsymbol{y}) = y_1$ and record positive crossings of the plane $h = 0$

$$\Sigma = \{\boldsymbol{y} : y_1 = 0\}, \qquad \Sigma^+ = \{\boldsymbol{y} \in \Sigma : -y_2 - y_3 > 0\}.$$

- Each recorded vector $\boldsymbol{q}_k \in \Sigma^+$ satisfies $h(\boldsymbol{q}_k) = 0$ and $\nabla h(\boldsymbol{q}_k) \cdot \boldsymbol{f}(\boldsymbol{q}_k) > 0$.
- Return map P sends each recorded crossing point to the next

$$\boldsymbol{q}_k \longmapsto \boldsymbol{q}_{k+1} = P(\boldsymbol{q}_k).$$

Each $\boldsymbol{q}_k$ lies in $\mathbb{R}^3$, but $y_1 = 0$ leaves two free coordinates. In section coordinates, P maps a subset of $\mathbb{R}^2$ into $\mathbb{R}^2$.

Hilltop Section $y_3' = 0$

- The third Rössler equation gives $y_3' = f_3(\boldsymbol{y}) = b + y_3(y_1 - c)$. Set $h(\boldsymbol{y}) = f_3(\boldsymbol{y})$. Record negative crossings of $h = 0$ in the **Hilltop Section**

$$\Sigma = \{\boldsymbol{y} : h(\boldsymbol{y}) = 0,\ y_3(-y_2 - y_3) < 0\}.$$

- Each recorded $\boldsymbol{q}_k \in \Sigma$ is a negative crossing since, on $h = 0$, $\nabla h \cdot \boldsymbol{f} = y_3(-y_2 - y_3) < 0$.
  - Each $\boldsymbol{q}_k$ marks a nondegenerate maximum of y_3 with $y_3' = 0$ and $y_3'' < 0$.
  - The section is curved because $h = 0$ gives $y_3(y_1 - c) = -b$, which links y_1 to y_3.
  - Each $\boldsymbol{q}_k$ lies in $\mathbb{R}^3$, but $h = 0$ leaves two free coordinates. In section coordinates, the return map sends a subset of $\mathbb{R}^2$ into $\mathbb{R}^2$.

Hilltop Return Map $y_3' = 0$

- Record $\boldsymbol{q}_k \in \Sigma$ at each selected maximum

$$\boldsymbol{q}_{k+1} = P_{\text{top}}(\boldsymbol{q}_k), \qquad P_{\text{top}} : D_{\text{top}} \subseteq \Sigma \longrightarrow \Sigma.$$

- For $\boldsymbol{y} \in \Sigma$, $b = 0.2$ gives

$$y_3 \neq 0, \qquad y_1 = c - \frac{b}{y_3}.$$

  The curved surface has two free coordinates.

- Peak height does not determine the next peak. Both section coordinates matter.

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Historical Cast of Characters

From planetary observations to chaotic flows

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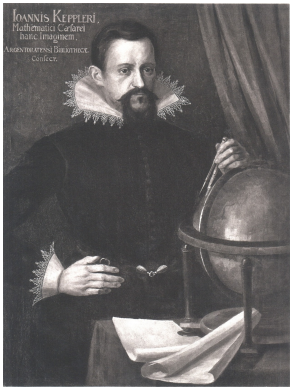
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Tycho Brahe (1546–1601)



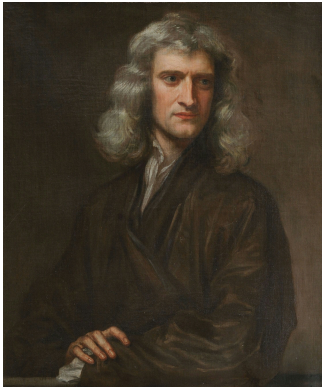
- Danish astronomer who made exceptionally precise observations before the telescope.
- Recorded the positions of Mars that Kepler later used to test planetary orbits.
- His measurements made small errors in circular models impossible to ignore.
- He also recorded the brilliant supernova of 1572.
- A duel cost him part of his nose; his prosthesis was likely brass.

Johannes Kepler (1571–1630)



- German astronomer and mathematician who worked with Brahe's observations.
- Found that Mars follows an ellipse, then formulated three laws of planetary motion.
- His laws described planetary motion accurately, setting a target for Newton's theory.
- An eight-arcminute mismatch in Mars's position helped convince him that circles would not do.
- He defended his mother when she was accused of witchcraft.

Isaac Newton (1642–1727)



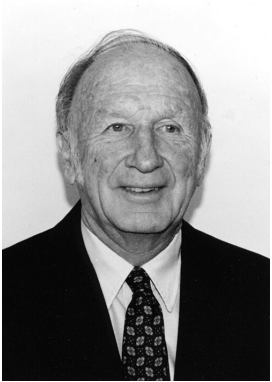
- English mathematician and physicist.
- In the 1687 *Principia*, gave laws of motion and universal gravitation.
- The same gravitational law explains falling bodies and Kepler's planetary orbits.
- He later ran the Royal Mint and pursued counterfeiters.
- He also kept extensive notes on alchemy.

Henri Poincaré (1854–1912)



- French mathematician who studied the geometry of trajectories without solving every equation explicitly.
- Investigated the three-body problem and the complex behavior of its orbits.
- Section crossings turn a continuous flow into a return map, the tool just used in this lecture.
- Correcting an error in his prize memoir revealed even more tangled three-body orbits.
- An insight once struck him as he stepped onto a carriage.

Edward Lorenz (1917–2008)



- American meteorologist at MIT who studied weather prediction with simplified equations.
- Showed that a deterministic model can amplify tiny differences in initial conditions.
- His 1963 work became a central example of chaos and limits to long-range prediction.
- A rerun with rounded input numbers produced a very different forecast.
- His famous 1972 talk asked whether a butterfly in Brazil could stir up a tornado in Texas.

Otto Rössler (born 1940)



- German scientist who proposed a compact three-variable chaotic flow in 1976.
- Its folded motion gives a clear setting for studying return maps.
- The Rössler flow supplied the plane and hilltop sections earlier in this lecture.
- His three-equation model needs only one nonlinear term.
- A taffy-pulling machine inspired its stretch-and-fold picture.

Summary

- Plane and hilltop sections record crossings of the Rössler flow.
- The return map sends each crossing point to the next.
- The historical cast connects planetary motion, differential equations, and chaotic flows.

History sources

- Brahe: NASA Earth Observatory on planetary observations, NASA on the 1572 supernova and his duel, and Kučera et al. on his prosthesis [7, 9, 8, 3].
- Kepler: NASA Earth Observatory and Owen Gingerich's biographical study, and the Stanford Encyclopedia of Philosophy [7, 2, 1].
- Newton: NASA Earth Observatory and The Royal Mint and the Newton Project's alchemical papers [7, 15, 14].
- Poincaré: the 1890 memoir and the King Oscar II prize history and Poincaré's *Science and Method* [10, 12, 11].
- Lorenz: MIT News on his work and the rounded-input rerun and his 1972 butterfly talk [5, 6].
- Rössler: *An Equation for Continuous Chaos* (1976) and Niklas Manz on the taffy machine [13, 4].

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